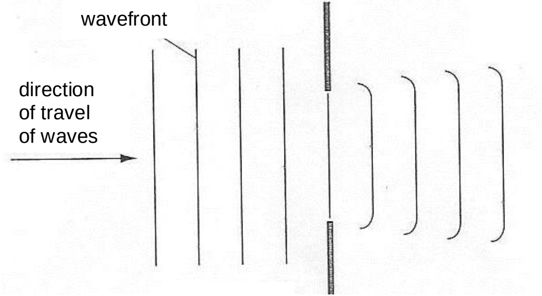
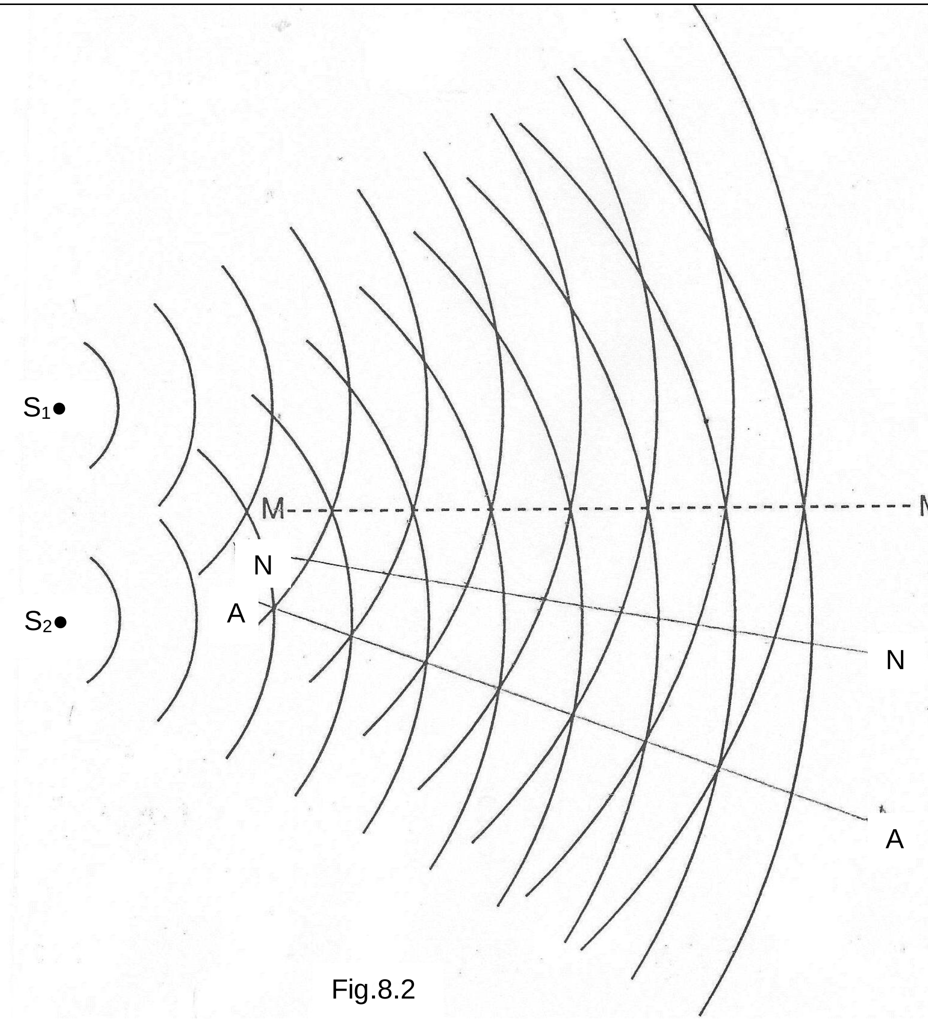
	Decrease in energy = ½ (32 x 10^{-3}) $4\pi^2$ $(2.0)^2$ [$(1.50 \times 10^{-2})^2$ – $(0.85 \times 10^{-2})^2$] = 3.9 x 10^{-4}J
8(a)(i)	 Fig.8.1
8(a)(ii)	When the slit is made narrower, the diffracted wavefronts will become more circular. The diffraction of waves will be more pronounced as the angle of diffraction will be much greater.
8(b)(i)	Coherent sources emit waves with a constant phase difference in which waves are of the same type with the same speed, frquency and wavelength.

8(b)(ii)	 Fig.8.2
8(c)(i)	$\lambda = \frac{ax}{D}$ $\lambda = \frac{(1.20X10^{-3})(1.3X10^{-3})}{247X10^{-2}}$ $= 6.3 \text{ X } 10^{-7}$ m
8(c)(ii)	Slit separation *a* must be much smaller than the slit to screen distance *D*, so that small angle approximation of $\tan\theta \approx \sin\theta$ used in the Young's double-slit formula can be valid.
8(c)(iii)1.	Contrast between bright and dark fringes will be reduced.The bright fringes will be less bright and dark fringes will not be completely dark due to incomplete destructive interference. However the fringe separations remain unchanged.
8(c)(iii)2.	More fringes will be observed due to greater diffraction of waves from narrower slits. The bright fringes will be less bright as less light can pass through the slits but the dark fringes will remain dark. The fringe separations will remain unchanged.

8(c)(iii)3.	The fringe separation x is directly proportional to the distance between the slits and screen D as wavelength and slit separation are kept constant. Hence the fringe separation will be smaller on the part of the screen that is nearer the slits and larger on the part of the screen that is further from the slits.The bright fringes on the screen nearer the slits will be brighter and there will be a greater contrast between bright and dark fringes whereas for fringes further from the slits, the bright fringes are dimmer and the contrast between bright and dark fringes will be reduced due to the attenuation of light.